

Amal Panwar

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Professional Summary

Data Scientist and Analytics professional with 7+ years of experience in machine learning, statistical modelling, AI solutions, and large-scale data analytics, along with a Master's degree in Data Analytics from the University of Strathclyde. Skilled in Python, SQL, R, Power BI, Azure, and big data technologies, with hands-on experience in predictive analytics, computer vision, RAG pipelines, and time series forecasting. Experienced in building scalable data pipelines, probabilistic models, and AI-driven systems that support data-driven decision-making. Passionate about leveraging advanced analytics and technology to solve complex business and sustainability challenges.

Professional Experience

Sporting Solutions

London, United Kingdom

Data Scientist - Quant

March 2025 – March 2026

- Improved **NFL odds across multiple markets** by analysing bet flow data and customer behaviour, identifying weaknesses in existing models, and refining pricing to reduce client losses.
- Built **ML pipeline in Python and statistical model in R** to predict outcomes across sports, improving pricing accuracy and competitive positioning.
- Developed betting markets using **F# and C#**, applying statistical methodologies and simulation techniques to model player and team performance.
- Created and maintained **pricing grids for various sports**, enabling consistent modelling of probabilities and odds across markets.
- Conducted **distributional and regression analysis and AB Testing** of player prop markets, identifying appropriate statistical fits for simulation and pricing.

Elgin City FC

Elgin, United Kingdom-Remote

Data Analyst/Scientist -Sports

June 2024 – Feb 2025

- Leveraged **WyScout** data and applied machine learning algorithms to identify high-potential player replacements, aiming to significantly enhance team performance.
- Conducted match analysis using **Computer Vision** techniques to support post-match evaluations, driving targeted improvements for future games.
- Created visual comparisons of target players against league benchmarks using **gauge charts** and **radar plots**, streamlining the scouting process and improving efficiency.
- Integrated data from diverse scouting reports to produce visuals that highlight player strengths and weaknesses, contributing to an anticipated **20%** improvement in team decision-making.
- Developed an **Agentic AI chatbot** using **RAG and LangChain** to help scouts interact with dashboards and retrieve actionable insights.

Caribbean Goods

Glasgow, United Kingdom

Data/Statistical Analyst

May 2023 – Dec 2023

- Researched environmental factors affecting coffee sustainability in Brazil, Guatemala, Honduras, and Nicaragua using data mining techniques (KDD) to identify hidden patterns and actionable insights.
- Analysed **macroeconomic** indicators to assess their influence on **Coffee Future** prices.
- Achieved a mean absolute percentage error of less than **7%** in price predictions using **statistical** and **econometric modelling**.
- Leveraged digital transformation and asset management strategies to help farmers optimise resource allocation, resulting in a **25%** increase in sustainable coffee growth in tropical regions.

Capgemini

Bengaluru, India

Data Analytics - Consultant

March 2022 – Aug 2022

- Extract and transform data in **Power BI** to fulfil a range of requests, creating insightful reports and identifying KPIs that align with business goals.

- Utilised the **Pytest** automation framework to validate **ETL** test cases and reduce **30%** testing time.

Cognizant

Bengaluru, India

Junior Data Engineer/Analyst

Nov 2017 – March 2022

- Spearheaded the development and implementation of a robust data warehouse framework, refining end-to-end data extraction, transformation, and loading (**ETL**) processes.
- Leveraged **PySpark** and **Azure Databricks** for scalable big data processing, reducing data pipeline execution time by **35%** and enhancing analytics capabilities.
- Utilised **T-SQL**, **MSEXcel**, and **Power BI** to uncover valuable insights, detect patterns and identify trends within complex datasets.
- Executed Data Quality Analysis, identifying anomalies to enhance overall data quality by **20% – 30%**.
- Utilised machine learning modelling techniques in **Python** to develop a **demand forecasting** model, leading to a **15%** reduction in inventory write-offs.
- Thrived in collaborative team environments, demonstrating a proactive approach, strong multitasking abilities, and a history of autonomous contributions that significantly elevated data analysis and reporting processes.

Skills

Programming: Python (NumPy, matplotlib, pandas, Scikit-learn), SQL, R (Shiny, Statistical Packages, ggplot2, rvest), Java, C/C++, DAX.

Skills: Mathematics (statistics and probability), Data Visualisation, Data mining, Microsoft Azure, Agile methodology, VSTS, SSIS/SSRS/SSAS, AI, Deep Learning, YOLO, openCV, RAG, LangChain, LLM, Time series analysis, Statistical analysis (Hypothesis testing, AB testing, p-value), Problem-solving and Risk Management, Digital Transformation.

Domain: Insurance, Retail and Consumer Goods, Financial, Health, and Sports.

Tools and Technology: Microsoft Office Suite (Dynamic 365, Power BI, Excel, etc.), SQL Server Management Studio, Jira, Guidewire, NetLogo, RStudio, Visual Studio Enterprise, Teradata, Postman, Jenkins, Git, Jupyter, and Visio.

Education

University of Strathclyde

Glasgow, UK

Master of Science in Data Analytics.

Sep 2022 – Sep 2023

- Award: Distinction
- Relevant coursework: Machine Learning, Data Science and Analytics in R, Big Data, Stochastic Modelling and Financial Econometrics.

Galgotias University

Greater Noida, India

Btech. Computer Science and Engineering.

Sep 2013 – Sep 2017

- GPA: 8.02/10.
- Relevant coursework: Computer programming and problem-solving, Object-oriented programming, Data structure and algorithm, Database basics, Software development, Data mining and warehousing, Artificial Intelligence.

References available on request:

- Liam Stewart (Elgin City FC, UK)
- Javier Gutierrez (Caribbean Goods, UK)
- Sandeep M (Cognizant, US)